```python
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.preprocessing import MultiLabelBinarizer

anime_dataset = pd.read_csv("./anime-dataset-2023.csv")
user_dataset = pd.read_csv("./users-details-2023.csv")
score_dataset = pd.read_csv("./users-score-2023.csv")

anime_dataset.head(5)
```

```
   anime_id                              Name             English name
\
0         1                      Cowboy Bebop             Cowboy Bebop

1         5  Cowboy Bebop: Tengoku no Tobira  Cowboy Bebop: The Movie

2         6                            Trigun                   Trigun

3         7                Witch Hunter Robin       Witch Hunter Robin

4         8                    Bouken Ou Beet   Beet the Vandel Buster


                         Other name Score  \
0                        カウボーイビバップ 8.75
1                   カウボーイビバップ 天国の扉 8.38
2                             トライガン 8.22
3  Witch Hunter ROBIN (ウイッチハンターロビン)  7.25
4                            冒険王ビィト 6.94

                                 Genres  \
0         Action, Award Winning, Sci-Fi
1                       Action, Sci-Fi
2            Action, Adventure, Sci-Fi
3  Action, Drama, Mystery, Supernatural
4      Adventure, Fantasy, Supernatural

                                            Synopsis   Type
Episodes  \
0  Crime is timeless. By the year 2071, humanity ...     TV     26.0

1  Another day, another bounty—such is the life o...  Movie      1.0

2  Vash the Stampede is the man with a $$60,000,0...     TV     26.0

3  Robin Sena is a powerful craft user drafted in...     TV     26.0

4  It is the dark century and the people are suff...     TV     52.0
```

```
                           Aired  ...          Studios    Source
Duration  \
0   Apr 3, 1998 to Apr 24, 1999  ...         Sunrise  Original  24 min
per ep
1                   Sep 1, 2001  ...           Bones  Original    1 hr
55 min
2   Apr 1, 1998 to Sep 30, 1998  ...        Madhouse     Manga  24 min
per ep
3   Jul 3, 2002 to Dec 25, 2002  ...         Sunrise  Original  25 min
per ep
4  Sep 30, 2004 to Sep 29, 2005  ...  Toei Animation     Manga  23 min
per ep

                            Rating    Rank Popularity Favorites Scored
By  \
0  R - 17+ (violence & profanity)    41.0         43     78525
914193.0
1  R - 17+ (violence & profanity)   189.0        602      1448
206248.0
2       PG-13 - Teens 13 or older   328.0        246     15035
356739.0
3       PG-13 - Teens 13 or older  2764.0       1795       613
42829.0
4                   PG - Children  4240.0       5126        14
6413.0

   Members                                            Image URL
0  1771505  https://cdn.myanimelist.net/images/anime/4/196...
1   360978  https://cdn.myanimelist.net/images/anime/1439/...
2   727252  https://cdn.myanimelist.net/images/anime/7/203...
3   111931  https://cdn.myanimelist.net/images/anime/10/19...
4    15001  https://cdn.myanimelist.net/images/anime/7/215...

[5 rows x 24 columns]
```

```
anime_dataset.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 24905 entries, 0 to 24904
Data columns (total 24 columns):
 #   Column        Non-Null Count  Dtype
---  ------        --------------  -----
 0   anime_id      24905 non-null  int64
 1   Name          24905 non-null  object
 2   English name  24905 non-null  object
 3   Other name    24905 non-null  object
 4   Score         24905 non-null  object
 5   Genres        24905 non-null  object
 6   Synopsis      24905 non-null  object
```

```
 7   Type           24905 non-null  object
 8   Episodes       24905 non-null  object
 9   Aired          24905 non-null  object
 10  Premiered      24905 non-null  object
 11  Status         24905 non-null  object
 12  Producers      24905 non-null  object
 13  Licensors      24905 non-null  object
 14  Studios        24905 non-null  object
 15  Source         24905 non-null  object
 16  Duration       24905 non-null  object
 17  Rating         24905 non-null  object
 18  Rank           24905 non-null  object
 19  Popularity     24905 non-null  int64
 20  Favorites      24905 non-null  int64
 21  Scored By      24905 non-null  object
 22  Members        24905 non-null  int64
 23  Image URL      24905 non-null  object
dtypes: int64(4), object(20)
memory usage: 4.6+ MB
```

```
anime_dataset.isna().sum()
```

```
anime_id         0
Name             0
English name     0
Other name       0
Score            0
Genres           0
Synopsis         0
Type             0
Episodes         0
Aired            0
Premiered        0
Status           0
Producers        0
Licensors        0
Studios          0
Source           0
Duration         0
Rating           0
Rank             0
Popularity       0
Favorites        0
Scored By        0
Members          0
Image URL        0
dtype: int64
```

```
anime_dataset.describe()
```

```
           anime_id    Popularity      Favorites       Members
count  24905.000000  24905.000000   24905.000000  2.490500e+04
mean   29776.709014  12265.388356     432.595222  3.710496e+04
std    17976.076290   7187.428393    4353.181647  1.568252e+05
min        1.000000      0.000000       0.000000  0.000000e+00
25%    10507.000000   6040.000000       0.000000  2.090000e+02
50%    34628.000000  12265.000000       1.000000  1.056000e+03
75%    45240.000000  18491.000000      18.000000  9.326000e+03
max    55735.000000  24723.000000  217606.000000  3.744541e+06
```

```
user_dataset.head(5)
```

```
   Mal ID Username  Gender                   Birthday
Location  \
0       1    Xinil    Male  1985-03-04T00:00:00+00:00
California
1       3  Aokaado    Male                        NaN          Oslo,
Norway
2       4  Crystal  Female                        NaN  Melbourne,
Australia
3       9   Arcane     NaN                        NaN
NaN
4      18      Mad     NaN                        NaN
NaN

                      Joined  Days Watched  Mean Score  Watching
Completed  \
0  2004-11-05T00:00:00+00:00         142.3        7.37       1.0
233.0
1  2004-11-11T00:00:00+00:00          68.6        7.34      23.0
137.0
2  2004-11-13T00:00:00+00:00         212.8        6.68      16.0
636.0
3  2004-12-05T00:00:00+00:00          30.0        7.71       5.0
54.0
4  2005-01-03T00:00:00+00:00          52.0        6.27       1.0
114.0

   On Hold  Dropped  Plan to Watch  Total Entries  Rewatched  Episodes
Watched
0      8.0     93.0           64.0          399.0       60.0
8458.0
1     99.0     44.0           40.0          343.0       15.0
4072.0
2    303.0      0.0           45.0         1000.0       10.0
12781.0
3      4.0      3.0            0.0           66.0        0.0
1817.0
4     10.0      5.0           23.0          153.0       42.0
3038.0
```

```
user_dataset.isna().sum()
```

```
Mal ID                    0
Username                  1
Gender               506907
Birthday             563222
Location             578485
Joined                    0
Days Watched              8
Mean Score                8
Watching                  8
Completed                 8
On Hold                   8
Dropped                   8
Plan to Watch             8
Total Entries             8
Rewatched                 8
Episodes Watched          8
dtype: int64
```

```
user_dataset.describe()
```

```
             Mal ID   Days Watched     Mean Score       Watching  \
count  7.312900e+05  731282.000000  731282.000000  731282.000000
mean   5.070203e+05      24.180819       3.948018       4.765714
std    3.640147e+05     140.105073       4.137606      20.495890
min    1.000000e+00       0.000000       0.000000       0.000000
25%    2.011085e+05       0.000000       0.000000       0.000000
50%    4.251705e+05       0.200000       0.000000       0.000000
75%    7.753400e+05      24.800000       8.040000       4.000000
max    1.291097e+06  105338.600000     255.000000    4358.000000

           Completed        On Hold        Dropped  Plan to Watch  \
count  731282.000000  731282.000000  731282.000000  731282.000000
mean       65.953066       3.391615       4.565480      17.547893
std       186.633286      19.296913      34.915341      90.286927
min         0.000000       0.000000       0.000000       0.000000
25%         0.000000       0.000000       0.000000       0.000000
50%         0.000000       0.000000       0.000000       0.000000
75%        48.000000       1.000000       1.000000       5.000000
max     13226.000000    5167.000000   14341.000000   21804.000000

       Total Entries      Rewatched  Episodes Watched
count  731282.000000  731282.000000      7.312820e+05
mean       96.230147       4.443352      1.658828e+03
std       265.459220      29.693175      5.077168e+04
min         0.000000       0.000000      0.000000e+00
25%         0.000000       0.000000      0.000000e+00
50%         1.000000       0.000000      1.500000e+01
```

```
75%        74.000000       0.000000      1.489000e+03
max     24817.000000   13215.000000      3.376442e+07
```

```
score_dataset.head(5)
```

```
   user_id Username  anime_id             Anime Title  rating
0        1    Xinil        21               One Piece       9
1        1    Xinil        48             .hack//Sign       7
2        1    Xinil       320                  A Kite       5
3        1    Xinil        49        Aa! Megami-sama!       8
4        1    Xinil       304  Aa! Megami-sama! Movie       8
```

```
score_dataset.isna().sum()
```

```
user_id          0
Username       232
anime_id         0
Anime Title      0
rating           0
dtype: int64
```

```
# On observe quelques doublons:
anime_counts = anime_dataset["Name"].value_counts()

# Afficher tous les animés qui sont dans la dataset anime_data (en
tenant compte des doublons)
len(anime_counts)
```

```
24901
```

```
# Afficher tous les doublons qui sont dans le dataset des animés
anime_dataset[anime_dataset["Name"].isin(anime_counts.index[anime_coun
ts.gt(1)])].sort_values("Name")
```

```
       anime_id       Name             English name  \
24137     54618  Awakening                  UNKNOWN
24840     55658  Awakening                  UNKNOWN
14749     38328  Azur Lane  Azur Lane the Animation
24586     55351  Azur Lane                Azur Lane
3053       3473   Souseiki              The Genesis
24807     55610   Souseiki                  UNKNOWN
8375      21103     Utopia                  UNKNOWN
24781     55582     Utopia                  UNKNOWN

                            Other name    Score
Genres  \
24137                        Awakening  UNKNOWN
UNKNOWN
24840                        Awakening  UNKNOWN
UNKNOWN
14749             アズールレーン THE ANIMATION     6.28         Action,
```

```
Sci-Fi
24586                          アズールレーン UNKNOWN  Action, Slice of
Life
3053                               創世記  5.38                UNKNOWN

24807                          UNKNOWN  UNKNOWN
Fantasy
8375   true tears×花咲くいろは×TARITARI ユートピア  6.41
UNKNOWN
24781                             ユートピア UNKNOWN
UNKNOWN

                                                   Synopsis     Type
Episodes  \
24137           No description available for this anime.    Music
1.0
24840   Music video for the song Awakening by Ken Ishii.    Music
1.0
14749  When the "Sirens," an alien force with an arse...       TV
12.0
24586  Assorted commercials for the Azur Lane Mobile ...  Special
UNKNOWN
3053   This is a short film parodying the major Holly...    Movie
1.0
24807  As Shoko Asahara is depicted in a less embelli...      OVA
1.0
8375   Utopia is a song that was specially created fo...    Music
1.0
24781           No description available for this anime.    Music
1.0

                             Aired  ...                    Studios  \
24137                 May 11, 2020  ...                    UNKNOWN
24840                 Oct 30, 2002  ...                    UNKNOWN
14749  Oct 3, 2019 to Mar 20, 2020  ...  Bibury Animation Studios
24586            Apr 17, 2020 to ?  ...                    UNKNOWN
3053                  Oct 1, 1968  ...                    UNKNOWN
24807                Not available  ...                    UNKNOWN
8375                  Feb 26, 2014  ...                 P.A. Works
24781                 Aug 12, 2021  ...                    UNKNOWN

            Source       Duration                      Rating     Rank
\
24137  Mixed media          3 min                     UNKNOWN      0.0

24840     Original          4 min               G - All Ages  UNKNOWN

14749         Game  23 min per ep           R+ - Mild Nudity   7502.0

24586         Game         30 sec  PG-13 - Teens 13 or older      0.0
```

```
3053       Unknown          4 min              PG - Children  11141.0

24807     Original        Unknown               G - All Ages      0.0

8375      Original          1 min  PG-13 - Teens 13 or older  UNKNOWN

24781     Original          3 min               G - All Ages      0.0


      Popularity Favorites Scored By Members  \
24137          0         0   UNKNOWN       0
24840      24580         0   UNKNOWN      35
14749       1343       960   58398.0  162902
24586          0         0   UNKNOWN       0
3053       10929         1     965.0    1831
24807          0         0   UNKNOWN       0
8375        6874         4    2740.0    7252
24781          0         0   UNKNOWN       0

                                                 Image URL
24137  https://cdn.myanimelist.net/images/anime/1535/...
24840  https://cdn.myanimelist.net/images/anime/1632/...
14749  https://cdn.myanimelist.net/images/anime/1106/...
24586  https://cdn.myanimelist.net/images/anime/1421/...
3053   https://cdn.myanimelist.net/images/anime/10/57...
24807  https://cdn.myanimelist.net/images/anime/1704/...
8375   https://cdn.myanimelist.net/images/anime/13/55...
24781  https://cdn.myanimelist.net/images/anime/1475/...

[8 rows x 24 columns]
```

```python
# On choisit de supprimer les doublons qui donnent le moins
d'informations
anime_ids_delete = [54618, 55351, 55610, 55582]

# Définir une fonction pour supprimer les doublons qui prend en entrée
un dataframe et une liste de ID
def remove_duplicates(df, anime_ids_delete):
  return df[ ~ df["anime_id"].isin(anime_ids_delete) ]

anime_dataset = remove_duplicates(anime_dataset, anime_ids_delete)
score_dataset = remove_duplicates(score_dataset, anime_ids_delete)

anime_dataset.value_counts().sort_values()
```

```
anime_id  Name                                    English name
Other name                  Score    Genres
Synopsis
Type     Episodes  Aired                          Premiered    Status
Producers                                              Licensors
```

Studios Source Duration Rating
Rank Popularity Favorites Scored By Members Image URL

1 Cowboy Bebop Cowboy Bebop
カウボーイビバップ 8.75 Action, Award Winning, Sci-Fi Crime
is timeless. By the year 2071, humanity has expanded across the galaxy, filling the surface of other planets with settlements like those on Earth. These new societies are plagued by murder, drug use, and theft, and intergalactic outlaws are hunted by a growing number of tough bounty hunters.\n\nSpike Spiegel and Jet Black pursue criminals throughout space to make a humble living. Beneath his goofy and aloof demeanor, Spike is haunted by the weight of his violent past. Meanwhile, Jet manages his own troubled memories while taking care of Spike and the Bebop, their ship. The duo is joined by the beautiful con artist Faye Valentine, odd child Edward Wong Hau Pepelu Tivrusky IV, and Ein, a bioengineered Welsh Corgi.\n\nWhile developing bonds and working to catch a colorful cast of criminals, the Bebop crew's lives are disrupted by a menace from Spike's past. As a rival's maniacal plot continues to unravel, Spike must choose between life with his newfound family or revenge for his old wounds.
TV 26.0 Apr 3, 1998 to Apr 24, 1999 spring 1998 Finished Airing Bandai Visual
Funimation, Bandai Entertainment Sunrise Original 24 min per ep R - 17+ (violence & profanity) 41.0 43 78525
914193.0 1771505
https://cdn.myanimelist.net/images/anime/4/19644.jpg 1
8 Bouken Ou Beet Beet the Vandel Buster
冒険王ビィト 6.94 Adventure, Fantasy, Supernatural
It is the dark century and the people are suffering under the rule of the devil, Vandel, who is able to manipulate monsters. The Vandel Busters are a group of people who hunt these devils, and among them, the Zenon Squad is known to be the strongest busters on the continent. A young boy, Beet, dreams of joining the Zenon Squad. However, one day, as a result of Beet's fault, the Zenon squad was defeated by the devil, Beltose. The five dying busters sacrificed their life power into their five weapons, Saiga. After giving their weapons to Beet, they passed away. Years have passed since then and the young Vandel Buster, Beet, begins his adventure to carry out the Zenon Squad's will to put an end to the dark century.
TV 52.0 Sep 30, 2004 to Sep 29, 2005 fall 2004 Finished Airing TV Tokyo, Dentsu
Illumitoon Entertainment Toei Animation Manga 23 min per ep PG - Children 4240.0 5126 14
6413.0 15001
https://cdn.myanimelist.net/images/anime/7/21569.jpg 1
15 Eyeshield 21 UNKNOWN
アイシールド21 7.92 Sports
Shy, reserved, and small-statured, Deimon High School student Sena Kobayakawa is the perfect target for bullies. However, as a result of

running errands throughout his life, Sena has become agile and developed a skill for avoiding crowds of people. After the cunning Youichi Hiruma—captain of the Deimon Devil Bats football team—witnesses Sena's rapid legs in motion, he coerces the timid boy into joining his squad.\n\nAs Hiruma wants to conceal Sena's identity from other clubs, Sena is forced to hide under the visored helmet of "Eyeshield 21," a mysterious running back wearing the number 21 jersey. The legendary Eyeshield 21 can supposedly run at the speed of light and has achieved remarkable feats in the United States during his time at the Notre Dame College.\n\nAccustomed to avoiding his problems in the past, Sena's specialty might just help him become the new secret weapon of the Deimon Devil Bats. As he interacts with his teammates, Sena gradually gains more self-confidence and forges valuable bonds along the way. TV 145.0 Apr 6, 2005 to Mar 19, 2008 spring 2005 Finished Airing TV Tokyo, Nihon Ad Systems, TV Tokyo Music, Shueisha VIZ Media, Sentai Filmworks Gallop Manga 23 min per ep PG-13 - Teens 13 or older 688.0 1252 1997 86524.0 177688 https://cdn.myanimelist.net/images/anime/1079/133529.jpg 1 16 Hachimitsu to Clover Honey and Clover ハチミツとクローバー 8.0 Comedy, Drama, Romance Yuuta Takemoto, a sophomore at an arts college, shares a cheap apartment with two seniors—the eccentric Shinobu Morita, who keeps failing to graduate due to his absenteeism, and the sensible Takumi Mayama, who acts as a proper senior to Takemoto, often looking out for him.\n\nTakemoto had not given much thought to his future until one fine spring day, when he meets the endearing Hagumi Hanamoto and falls in love at first sight. Incredibly gifted in the arts, Hagumi enrolls in Takemoto's university and soon befriends the popular pottery student Ayumi Yamada. Ayumi is already well acquainted with the three flatmates and secretly harbors deep feelings for one of them.\n\nHachimitsu to Clover is a heartwarming tale of youth, love, soul-searching, and self-discovery, intricately woven through the complex relationships between five dear friends. TV 24.0 Apr 15, 2005 to Sep 27, 2005 spring 2005 Finished Airing Dentsu, Genco, Fuji TV, Asmik Ace, Shueisha VIZ Media, Discotek Media J.C.Staff Manga 23 min per ep PG-13 - Teens 13 or older 589.0 862 4136 81747.0 260166 https://cdn.myanimelist.net/images/anime/1301/133577.jpg 1 17 Hungry Heart: Wild Striker UNKNOWN ハングリーハートWild Striker 7.55 Comedy, Slice of Life, Sports As the younger brother of Japanese soccer star Seisuke Kanou, Kyousuke was always expected to grow as a soccer player at the same pace his brother did—an expectation that proved too difficult to meet. Having fallen behind, he now lives in the shadow of his brother's success.\n\nEntering his freshman year at Jouyou Akanegaoka High School, Kyousuke vows never to play soccer again. However, Miki Tsujiwaki, the captain of the girls' soccer team, and Mori Kazuto, the

manager of the boys' team, recognize Kyousuke's potential and want to see his return to the game for their own reasons.\n\nWith an opportunity to play soccer again, Kyousuke must either remain steadfast in his decision to abandon the sport he once loved, or allow himself to reignite that flame to become the best striker in the world.
TV 52.0 Sep 11, 2002 to Sep 10, 2003 fall 2002 Finished Airing UNKNOWN UNKNOWN Nippon Animation Manga 23 min per ep PG-13 - Teens 13 or older 1551.0 4212 237 12960.0 24172 https://cdn.myanimelist.net/images/anime/12/49655.jpg 1

..
55707 Kokoro UNKNOWN 心 UNKNOWN UNKNOWN Music video for the song Kokoro by Shikidrop.
Music 1.0 Jul 30, 2020 UNKNOWN Finished Airing UNKNOWN UNKNOWN UNKNOWN Original 4 min PG-13 - Teens 13 or older UNKNOWN 24664 0 UNKNOWN 31 https://cdn.myanimelist.net/images/anime/1146/136550.jpg 1
55706 Music UNKNOWN ミュージック UNKNOWN UNKNOWN Music video for the song Music by Kuusou Iinkai.
Music 1.0 Jan 22, 2016 UNKNOWN Finished Airing UNKNOWN UNKNOWN UNKNOWN Original 4 min G - All Ages UNKNOWN 24671 0 UNKNOWN 30 https://cdn.myanimelist.net/images/anime/1779/136548.jpg 1
55678 Twilight (#1 Abstract na Ringo-tachi) UNKNOWN Twilight (#1 アブストラクトな林檎たち) UNKNOWN Avant Garde Music video for the song Twilight (#1 Abstract na Ringo-tachi) by Denki Groove and Scha Dara Parr.
Music 1.0 Apr 27, 2005 UNKNOWN Finished Airing UNKNOWN UNKNOWN UNKNOWN Original 4 min PG-13 - Teens 13 or older UNKNOWN 24636 0 UNKNOWN 31 https://cdn.myanimelist.net/images/anime/1637/136488.jpg 1
55716 Mechronicle Mechronicle Mechronicle UNKNOWN UNKNOWN Set in a mechanical world, Mechronicle tells the story of Mechanico: a small, shy, withdrawn boy with big glasses. He is totally in love with Nell, who looks after Mechanico like he's her kid brother. One day, Nell is kidnapped by Team Magu Magu, a villainous group of machines. She soon discovers that the secret to their world is hidden in her machinery. Mechanico vows to rescue Nell and, plucking up his courage, pursues the kidnappers. The adventure begins!\n\nSites:\nhttps://anime-tokyo.com/pdf/page/english/works-2017-en/mappa.pdf\n\nhttps://anime-tokyo.com/english/works-2017-en/

```
#mappa
UNKNOWN  UNKNOWN   Not available                UNKNOWN      Not yet
aired    UNKNOWN                                                UNKNOWN
UNKNOWN           Original  Unknown        UNKNOWN
0.0      0           0          UNKNOWN    0
https://cdn.myanimelist.net/images/anime/1074/136570.jpg    1
55735     Shijuuku Nichi                      UNKNOWN
四十九日         UNKNOWN  UNKNOWN                            A music
video for the song Shijuuku Nichi by Shannon.
Music    1.0       Sep 5, 2022                 UNKNOWN      Finished
Airing  UNKNOWN                                              UNKNOWN
UNKNOWN           Original  3 min          PG-13 - Teens 13 or older
0.0      0           0          UNKNOWN    0
https://cdn.myanimelist.net/images/anime/1902/136595.jpg    1
Name: count, Length: 24901, dtype: int64
```

```python
# Combien d'animés ?
# Combien d'utilisateurs?

nb_animes = len(anime_dataset)
print(f"Il y a {nb_animes} animés.")
```

```
Il y a 24901 animés.
```

```python
nb_users = len(score_dataset["user_id"].unique())
print(f"Il y a {nb_users} utilisateurs.")
```

```
Il y a 270033 utilisateurs.
```

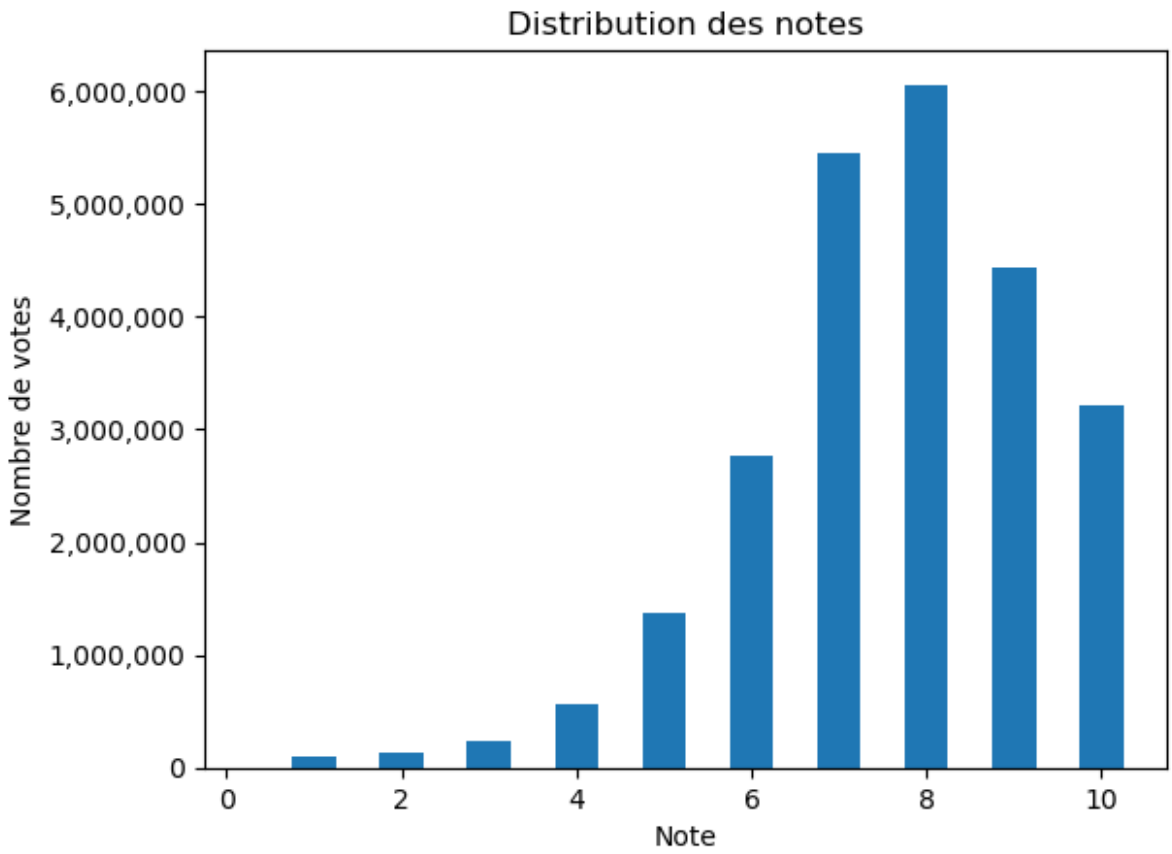
```python
# Créer la matrice des notes (une ligne par utilisateur, une colonne
par film). Quelle est la proportion de notes manquantes ?

# IndexError: index 159767137 is out of bounds for axis 0 with size
159767105
# r = score_dataset.pivot_table(index="user_id", columns="anime_id",
values="rating")
# nb_nan = r.isna().sum().sum()
# ratio_nan = nb_nan/r.size
# print(f"{ratio_nan*100:.2f}% des valeurs sont manquantes.")

##### Etudier la variable notes (paramètres statistiques et
représentations graphiques)

# La note moyenne
score_dataset["rating"].describe()
```

```
count    2.432491e+07
mean     7.622947e+00
std      1.661504e+00
min      1.000000e+00
25%      7.000000e+00
```

```
50%      8.000000e+00
75%      9.000000e+00
max      1.000000e+01
Name: rating, dtype: float64
```

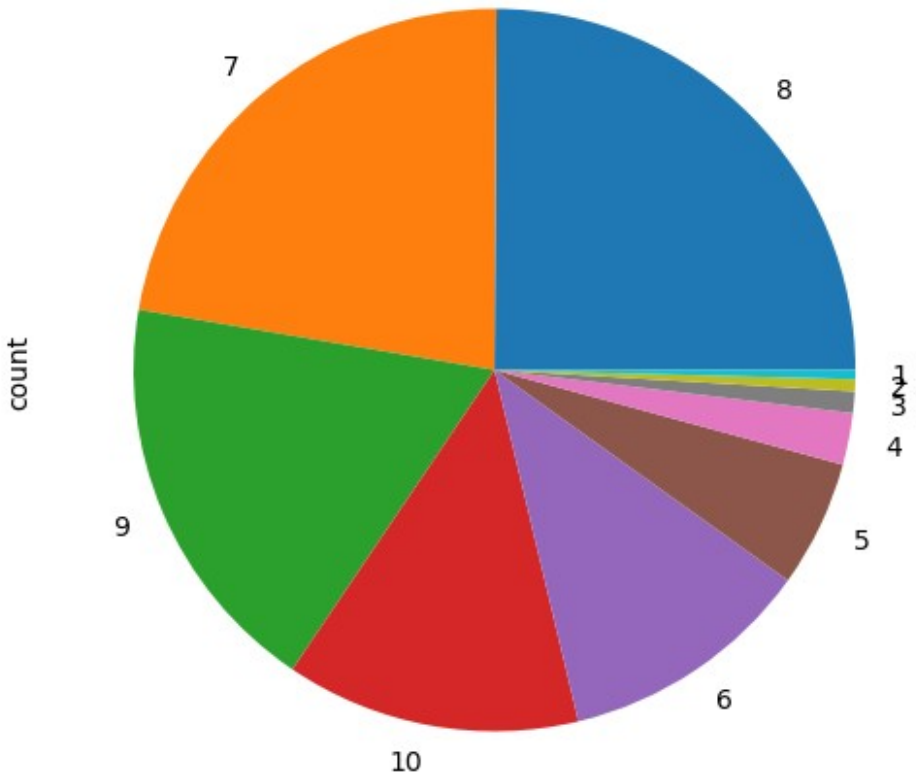
```
import matplotlib.ticker as mtick

# Histogramme des notes (avec un pas de 0.5) et format des ticks en
entiers
plt.figure(figsize=(10,6))
ax = score_dataset.hist("rating", bins=np.arange(0.25, 10.5, 0.5),
grid=False)

# Forcer l'affichage des fréquences sur l'axe Y avec des entiers
plt.gca().yaxis.set_major_formatter(mtick.FuncFormatter(lambda x, _:
f'{int(x):,}'))

plt.title('Distribution des notes')
plt.xlabel('Note')
plt.ylabel('Nombre de votes')
plt.show()
```

```
<Figure size 1000x600 with 0 Axes>
```



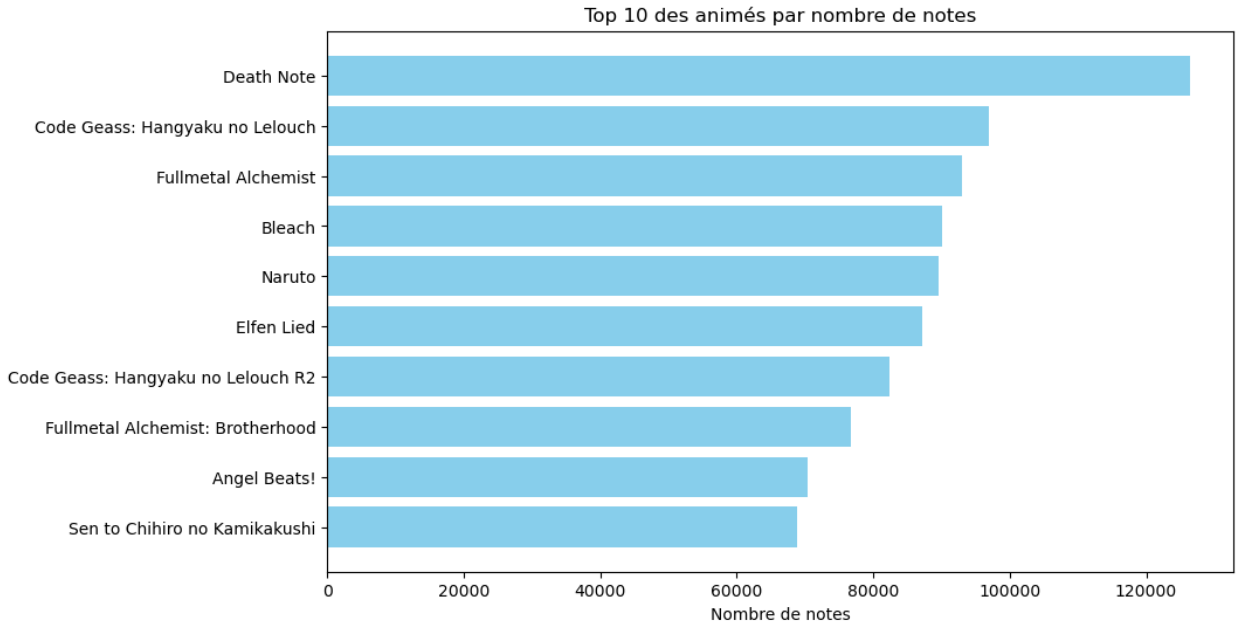
```
# Diagramme circulaire
plt.figure(figsize=(6,15))
score_dataset["rating"].value_counts().plot.pie()
```

```
<Axes: ylabel='count'>
```



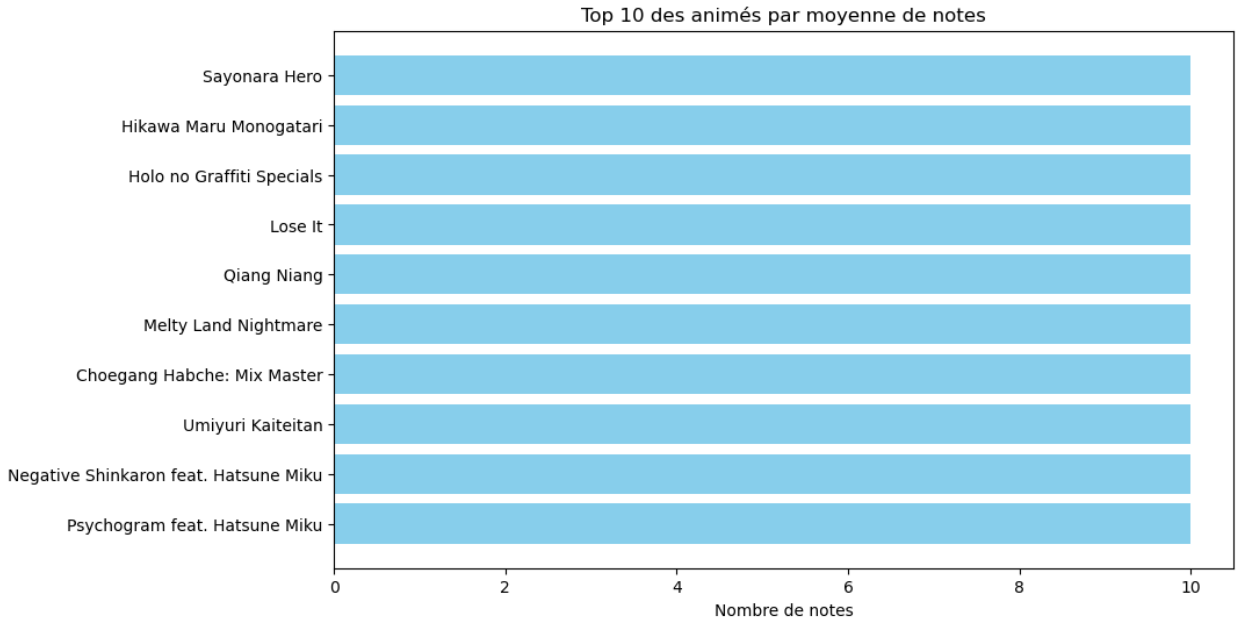

```
# Définir une variable égale au nombre de notes reçues par chaque
animé.
# Etudier cette variable (paramètres statistiques et représentations
graphiques).

dict_title_to_id = dict(zip( anime_dataset["Name"],
anime_dataset["anime_id"] ))
dict_id_to_title = {v : k for k, v in dict_title_to_id.items()}

nb_ratings_per_anime =
score_dataset.groupby(by="anime_id").size().sort_values(ascending=Fals
e)
nb_ratings_per_anime = nb_ratings_per_anime.rename("nb ratings")
```

```
nb_ratings_per_anime.index = nb_ratings_per_anime.index.map(lambda x:
dict_id_to_title.get(x, f"Unknown anime_id {x}"))

# Sélectionner les 10 premiers animés
top_10_anime = nb_ratings_per_anime.head(10)

# Créer un diagramme en barres
plt.figure(figsize=(10,6))
plt.barh(top_10_anime.index, top_10_anime.values, color='skyblue')
plt.xlabel('Nombre de notes')
plt.title('Top 10 des animés par nombre de notes')
plt.gca().invert_yaxis()  # Inverser l'axe des y pour avoir le 1er en
haut
plt.show()
```




```
# # Définir une variable égale à la note moyenne de chaque animé.
# Etudier cette variable (paramètres statistiques et représentations
graphiques).

# Calculate the mean rating for each anime_id
average_ratings_per_anime = score_dataset.groupby(by="anime_id")
["rating"].mean().sort_values(ascending=False)
average_ratings_per_anime = average_ratings_per_anime.rename("average
ratings")
average_ratings_per_anime =
average_ratings_per_anime[average_ratings_per_anime.index.isin(dict_id
_to_title.keys())]
average_ratings_per_anime.index =
average_ratings_per_anime.index.map(lambda x: dict_id_to_title[x])
```

```
# Sélectionner les 10 premiers animés
top_10_anime = average_ratings_per_anime.head(10)

# Créer un diagramme en barres
plt.figure(figsize=(10,6))
plt.barh(top_10_anime.index, top_10_anime.values, color='skyblue')
plt.xlabel('Nombre de notes')
plt.title('Top 10 des animés par moyenne de notes')
plt.gca().invert_yaxis()  # Inverser l'axe des y pour avoir le 1er en
haut
plt.show()
```

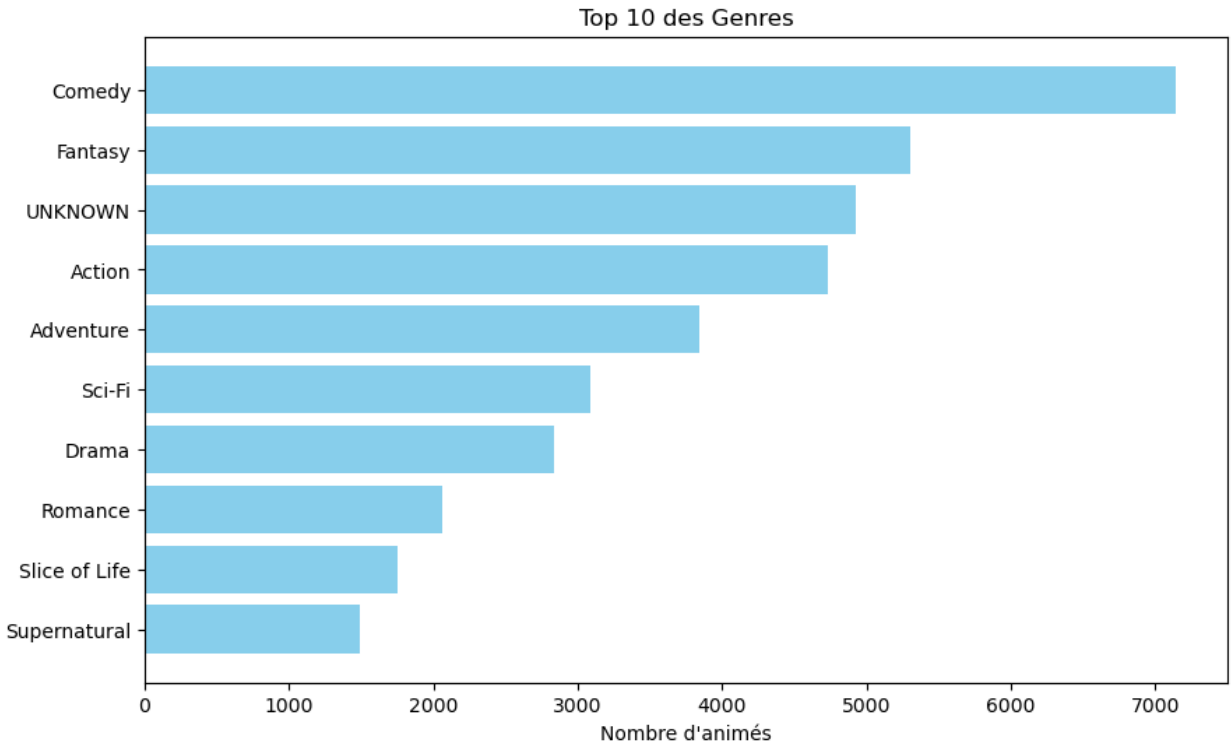



```
# Pas pertinent, il peut y avoir un seul utilisateur qui a attribué
une note 10/10

# Nombre d'animés par genre
list_genres = anime_dataset["Genres"].str.split(",
").explode().unique()

count_genres = pd.get_dummies(anime_dataset["Genres"].str.split(",
").explode()).sum().sort_values(ascending=False)
count_genres = count_genres.rename("count")

count_genres
```

```
Comedy            7142
Fantasy           5305
UNKNOWN           4927
Action            4729
Adventure         3842
```

```
Sci-Fi            3091
Drama             2836
Romance           2063
Slice of Life     1754
Supernatural      1494
Hentai            1486
Mystery            847
Avant Garde        804
Ecchi              795
Sports             771
Horror             534
Suspense           242
Award Winning      241
Boys Love          169
Gourmet            146
Girls Love         113
Erotica             54
Name: count, dtype: int64
```

```
# Sélectionner les 10 premiers animés
top_10_anime = count_genres.head(10)

# Créer un diagramme en barres
plt.figure(figsize=(10,6))
plt.barh(top_10_anime.index, top_10_anime.values, color='skyblue')
plt.xlabel('Nombre d\'animés')
plt.title('Top 10 des Genres')
plt.gca().invert_yaxis()  # Inverser l'axe des y pour avoir le 1er en
haut
plt.show()
```




```python
# Note moyenne de chaque genre

df_mean_rating_genres =
pd.get_dummies(anime_dataset["Genres"].str.split(", ").explode())
df_mean_rating_genres.index.name = "index"

df_mean_rating_genres
```

```
       Action  Adventure  Avant Garde  Award Winning  Boys Love
Comedy  \
index

0        True      False        False          False      False
False
0       False      False        False           True      False
False
0       False      False        False          False      False
False
1        True      False        False          False      False
False
1       False      False        False          False      False
False
...       ...        ...          ...            ...        ...     ..
.
24902   False       True        False          False      False
False
24902   False      False        False          False      False
```

```
False
24902   False      False        False          False      False
False
24903   False      False        False          False      False
False
24904   False      False        False          False      False
False

       Drama  Ecchi  Erotica  Fantasy  ...  Hentai  Horror  Mystery
Romance  \
index                                   ...

0      False  False    False    False  ...   False   False    False
False
0      False  False    False    False  ...   False   False    False
False
0      False  False    False    False  ...   False   False    False
False
1      False  False    False    False  ...   False   False    False
False
1      False  False    False    False  ...   False   False    False
False
...      ...    ...      ...      ...  ...     ...     ...      ...
...
24902  False  False    False    False  ...   False   False    False
False
24902  False  False    False     True  ...   False   False    False
False
24902  False  False    False    False  ...   False   False    False
False
24903  False  False    False    False  ...   False   False    False
False
24904  False  False    False    False  ...   False   False    False
False

       Sci-Fi  Slice of Life  Sports  Supernatural  Suspense  UNKNOWN

index

0       False          False   False         False     False    False

0       False          False   False         False     False    False

0        True          False   False         False     False    False

1       False          False   False         False     False    False

1        True          False   False         False     False    False

...       ...             ...     ...           ...       ...      ...
```

```
24902   False          False   False         False     False    False

24902   False          False   False         False     False    False

24902    True          False   False         False     False    False

24903   False          False   False         False     False     True

24904   False          False   False         False     False     True


[43385 rows x 22 columns]
```

```python
# One hot encoding des genres de chaque animé
df_mean_rating_genres = df_mean_rating_genres.groupby("index").sum()

df_mean_rating_genres
```

```
       Action  Adventure  Avant Garde  Award Winning  Boys Love
Comedy  \
index

0           1          0            0              1          0
0
1           1          0            0              0          0
0
2           1          1            0              0          0
0
3           1          0            0              0          0
0
4           0          1            0              0          0
0
...       ...        ...          ...            ...        ...     ..
.
24900       0          0            0              0          0
1
24901       1          1            0              0          0
0
24902       1          1            0              0          0
0
24903       0          0            0              0          0
0
24904       0          0            0              0          0
0

       Drama  Ecchi  Erotica  Fantasy  ...  Hentai  Horror  Mystery
Romance  \
index                                  ...

0          0      0        0        0  ...       0       0        0
```

```
0
1          0      0        0        0  ...       0       0        0
0
2          0      0        0        0  ...       0       0        0
0
3          1      0        0        0  ...       0       0        1
0
4          0      0        0        1  ...       0       0        0
0
...      ...    ...      ...      ...  ...     ...     ...      ...
...
24900      0      0        0        1  ...       0       0        0
0
24901      0      0        0        1  ...       0       0        0
0
24902      0      0        0        1  ...       0       0        0
0
24903      0      0        0        0  ...       0       0        0
0
24904      0      0        0        0  ...       0       0        0
0

       Sci-Fi  Slice of Life  Sports  Supernatural  Suspense  UNKNOWN

index

0           1              0       0             0         0        0

1           1              0       0             0         0        0

2           1              0       0             0         0        0

3           0              0       0             1         0        0

4           0              0       0             1         0        0

...       ...            ...     ...           ...       ...      ...

24900       0              1       0             0         0        0

24901       0              0       0             0         0        0

24902       1              0       0             0         0        0

24903       0              0       0             0         0        1

24904       0              0       0             0         0        1


[24901 rows x 22 columns]
```

```
df_mean_rating_genres["anime_id"] = anime_dataset["anime_id"]

df_mean_rating_genres
```

```
       Action  Adventure  Avant Garde  Award Winning  Boys Love
Comedy  \
index

0           1          0            0              1          0
0
1           1          0            0              0          0
0
2           1          1            0              0          0
0
3           1          0            0              0          0
0
4           0          1            0              0          0
0
...       ...        ...          ...            ...        ...     ..
.
24900       0          0            0              0          0
1
24901       1          1            0              0          0
0
24902       1          1            0              0          0
0
24903       0          0            0              0          0
0
24904       0          0            0              0          0
0

       Drama  Ecchi  Erotica  Fantasy  ...  Horror  Mystery  Romance
Sci-Fi  \
index                                   ...

0          0      0        0        0  ...       0        0        0
1
1          0      0        0        0  ...       0        0        0
1
2          0      0        0        0  ...       0        0        0
1
3          1      0        0        0  ...       0        1        0
0
4          0      0        0        1  ...       0        0        0
0
...      ...    ...      ...      ...  ...     ...      ...      ...
...
24900      0      0        0        1  ...       0        0        0
0
24901      0      0        0        1  ...       0        0        0
```

```
0
24902      0      0        0        1  ...       0        0        0
1
24903      0      0        0        0  ...       0        0        0
0
24904      0      0        0        0  ...       0        0        0
0

       Slice of Life  Sports  Supernatural  Suspense  UNKNOWN
anime_id
index

0                  0       0             0         0        0
1
1                  0       0             0         0        0
5
2                  0       0             0         0        0
6
3                  0       0             1         0        0
7
4                  0       0             1         0        0
8
...              ...     ...           ...       ...      ...       ..
.
24900              1       0             0         0        0
55731
24901              0       0             0         0        0
55732
24902              0       0             0         0        0
55733
24903              0       0             0         0        1
55734
24904              0       0             0         0        1
55735

[24901 rows x 23 columns]
```

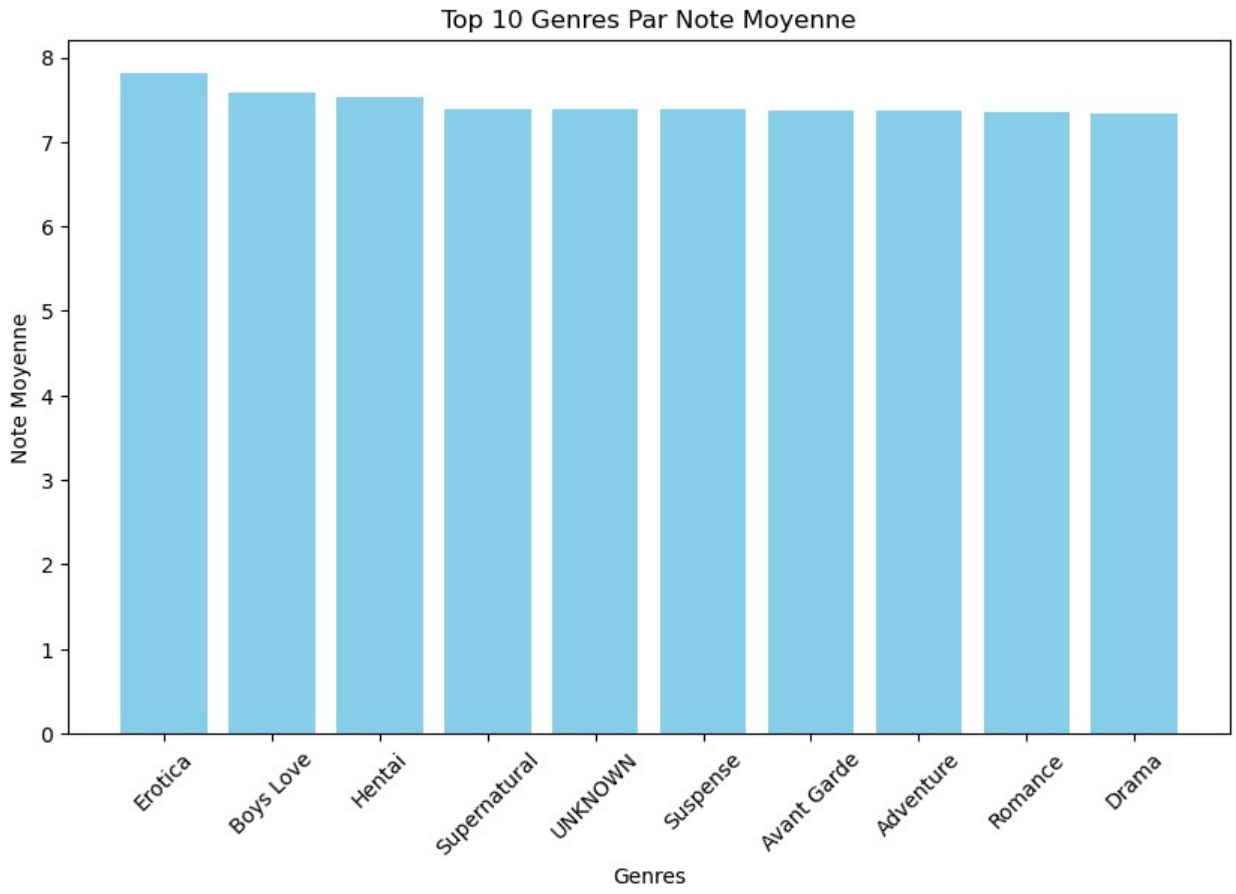
```python
# On réindexe selon le movieId
df_mean_rating_genres = df_mean_rating_genres.set_index("anime_id")

# On ne garde que les anime_id communs aux deux datasets
common_anime_ids =
df_mean_rating_genres.index.intersection(score_dataset["anime_id"])

# Filtrer df_mean_rating_genres selon ces anime_id
df_mean_rating_genres = df_mean_rating_genres.loc[common_anime_ids]

# On ajoute la colonne des notes pour chaque film
df_mean_rating_genres["rating"] = score_dataset["rating"]

df_mean_rating_genres
```

```
          Action  Adventure  Avant Garde  Award Winning  Boys Love
Comedy  \
anime_id

1              1          0            0              1          0
0
5              1          0            0              0          0
0
6              1          1            0              0          0
0
7              1          0            0              0          0
0
8              0          1            0              0          0
0
...          ...        ...          ...            ...        ...
...
55687          0          0            1              0          0
0
55688          0          0            0              0          0
0
55720          0          0            0              0          0
1
55731          0          0            0              0          0
1
55733          1          1            0              0          0
0

          Drama  Ecchi  Erotica  Fantasy  ...  Horror  Mystery
Romance  \
anime_id                                   ...

1             0      0        0        0  ...       0        0
0
5             0      0        0        0  ...       0        0
0
6             0      0        0        0  ...       0        0
0
7             1      0        0        0  ...       0        1
0
8             0      0        0        1  ...       0        0
0
...         ...    ...      ...      ...  ...     ...      ...       ..
.
55687         0      0        0        0  ...       0        0
0
55688         0      0        0        0  ...       0        0
0
55720         0      0        0        1  ...       0        0
0
55731         0      0        0        1  ...       0        0
```

```
0
55733         0      0        0        1  ...       0        0
0

           Sci-Fi  Slice of Life  Sports  Supernatural  Suspense
UNKNOWN  \
anime_id

1               1              0       0             0         0
0
5               1              0       0             0         0
0
6               1              0       0             0         0
0
7               0              0       0             1         0
0
8               0              0       0             1         0
0
...           ...            ...     ...           ...       ...       .
..
55687           0              0       0             0         0
0
55688           0              0       0             0         0
1
55720           0              0       0             0         0
0
55731           0              1       0             0         0
0
55733           1              0       0             0         0
0

           rating
anime_id
1             7.0
5             8.0
6             7.0
7             5.0
8             6.0
...           ...
55687         9.0
55688         7.0
55720         8.0
55731         8.0
55733         8.0

[16480 rows x 23 columns]
```

```
genre_ratings = {g : df_mean_rating_genres[df_mean_rating_genres[g] ==
1]["rating"].mean() for g in list_genres}
genre_ratings
```

```
{'Action': 7.312439024390244,
 'Award Winning': 7.259090909090909,
 'Sci-Fi': 7.318235525797558,
 'Adventure': 7.3658362989323845,
 'Drama': 7.336411609498681,
 'Mystery': 7.294573643410852,
 'Supernatural': 7.389230769230769,
 'Fantasy': 7.3349725195256,
 'Sports': 7.239932885906041,
 'Comedy': 7.293470149253731,
 'Romance': 7.347177848775293,
 'Slice of Life': 7.3296875,
 'Suspense': 7.381909547738694,
 'Ecchi': 7.265888456549935,
 'Gourmet': 7.278260869565218,
 'Avant Garde': 7.3785425101214575,
 'Horror': 7.199134199134199,
 'Girls Love': 7.179245283018868,
 'Boys Love': 7.588235294117647,
 'Hentai': 7.529692832764505,
 'UNKNOWN': 7.381985892566468,
 'Erotica': 7.813953488372093}

# Trier les genres par la note moyenne
sorted_genres = dict(sorted(genre_ratings.items(), key=lambda item:
item[1], reverse=True))

# Sélectionner les 10 premiers genres
top_10_genres = dict(list(sorted_genres.items())[:10])

# Tracer l'histogramme
plt.figure(figsize=(10,6))
plt.bar(top_10_genres.keys(), top_10_genres.values(), color='skyblue')
plt.xlabel('Genres')
plt.ylabel('Note Moyenne')
plt.title('Top 10 Genres Par Note Moyenne')
plt.xticks(rotation=45)
plt.show()
```


Top 10 Genres Par Note Moyenne
Note Moyenne
0
1
2
3
4
5
6
7
8
Erotica
Boys Love
Hentai
Supernatural
UNKNOWN
Suspense
Avant Garde
Adventure
Romance
Drama
Genres